



Unit 4 · Outcome 1 · Climate Change

Greenhouse Gas Warming Potential

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE BIG IDEA

01 A tonne of methane is not a tonne of carbon dioxide

At the Cape Wirreanda Climate Observatory — our fictional monitoring station on Victoria's far south coast, facing the Southern Ocean — a flux tower samples the air drifting in from the land behind it. On any given day it picks up carbon dioxide from a natural-gas peaking plant, methane from the dairy paddocks, nitrous oxide rising off fertilised soil, and a whisper of sulfur hexafluoride leaking from the electricity switchyard.

The observatory has a problem familiar to every climate scientist: these gases are not interchangeable. One kilogram of methane traps far more heat than one kilogram of carbon dioxide. One kilogram of sulfur hexafluoride traps tens of thousands of times more. You cannot just add up the tonnes and call it "emissions" — that would be like adding 3 apples to 3 elephants and reporting "6 things."

To compare gases fairly, and to add them into a single regional total, scientists need a common currency. That currency is greenhouse gas warming potential — usually written Global Warming Potential (GWP). This dot point is about what that number means, what makes it big or small, and the calculation examiners love to set with it.

◆ NOTE

■ VCAA key knowledge "Greenhouse gas warming potential — a measure of the infrared radiation a gas absorbs over its lifetime in the atmosphere."

WHAT THE NUMBER ACTUALLY MEANS

02 Defining Global Warming Potential

A measure of how much heat (infrared radiation) one kilogram of a greenhouse gas traps in the atmosphere over its lifetime, relative to one kilogram of carbon dioxide, over a set period of time (the time horizon).

Three things are doing the heavy lifting in that definition, and each one is worth marks:

⚠ WATCH OUT

Exam trap — "no units" GWP is a ratio, so it has no units. Writing "28 °C" or "28 watts" for methane's GWP loses the mark. It is simply 28 — methane traps 28 times as much heat as the same mass of CO₂ over 100 years.

THE TWO INGREDIENTS

03 What makes a gas's warming potential big or small

The VCAA wording — "the infrared radiation a gas absorbs over its lifetime" — quietly packs in the two factors that set GWP. A potent gas is one that is good at catching heat and sticks around to keep doing it.

Earth radiates heat upward as infrared (longwave) radiation. Some molecules are shaped so they absorb this radiation very strongly — they vibrate and re-radiate heat back, with each molecule acting like a tiny, very effective heat trap. Methane and the fluorinated industrial gases absorb far more strongly per molecule than CO₂ does.

A gas that is removed quickly (broken down, dissolved or absorbed) only traps heat for a short while. A gas that lingers for centuries keeps trapping heat the whole time. Methane breaks down in about 12 years; carbon dioxide effectively contributes for centuries; sulfur hexafluoride survives for over 3,000 years.

Multiply how hard it traps by how long it traps, compare to CO₂, and you have GWP. Drag the two sliders below to feel how each factor pushes the number.

THE TIME-HORIZON TWIST

04 Why methane's number changes — the sprinter and the marathon runner

◆ ANALOGY

◆ Analogy — sprinter vs marathon runner Methane is a sprinter: it traps heat intensely, then it's gone in ~12 years. Carbon dioxide is a marathon runner: modest per kilogram, but it just keeps going for centuries. Over a 20-year race, the sprinter's burst dominates — methane's GWP is around 80. Stretch the race to 100 years and the sprinter finished long ago while the marathoner is still running, so methane's averaged-out number falls to about 28.

This is why GWP is always tied to a time horizon. The same gas has different GWPs depending on how long a window you integrate over. Long-lived gases (CO₂, N₂O, SF₆) barely change between the 20- and 100-year figures; short-lived methane changes a lot. Drag the horizon below and watch the accumulated heat — the area under each curve — rebalance.

THE NUMBERS WORTH KNOWING

05 Warming potentials of the gases you'll meet

You will not be asked to memorise a long table, but you should know the order of magnitude and the pattern: CO₂ low but long-lived; methane in the tens; nitrous oxide in the hundreds; the fluorinated industrial gases in the thousands to tens of thousands. These are 100-year values; Australia reports using the IPCC Fifth Assessment Report figures.

Greenhouse gas	GWP (100 yr)	Lifetime	Main local source at Cape Wirreanda
Carbon dioxide CO ₂	1	centuries+	gas peaking plant, vehicles
Methane CH ₄	~28	~12 yr	dairy cattle, wetland, gas leaks
Nitrous oxide N ₂ O	~265	~115 yr	nitrogen fertiliser on soil
HFC-134a refrigerant	~1,300	~14 yr	cold-store & car air-con leaks
Sulfur hexafluoride SF ₆	~23,500	~3,200 yr	high-voltage switchgear



Figure · Greenhouse gas compared at a glance.

◆ **NOTE**

■ Why CO₂ is still public enemy number one CO₂ has the smallest GWP in the table — yet it causes the most warming. Hold that paradox; section 07 resolves it. (Hint: GWP is per kilogram, and we emit staggeringly more CO₂ than anything else.)

THE COMMON CURRENCY

06 Carbon dioxide equivalent (CO₂-e)

GWP lets us convert any gas into the amount of CO₂ that would cause the same warming. That converted figure is carbon dioxide equivalent (CO₂-e), and the formula is the simplest in the whole course:

◆ **NOTE**

Σ The formula examiners want CO₂-e (tonnes) = mass of gas (tonnes) × GWP

The Cape Wirreanda team needs one regional total. Set each source's annual emissions and watch the raw tonnes convert into a fair, comparable CO₂-e total. Notice how a small mass of a high-GWP gas can outweigh a large mass of a low-GWP one.

◆ **NOTE**

✓ Worked example — show your working The dairy emits 400 t of methane. In CO₂-e that is $400 \times 28 = 11,200$ t CO₂-e. The switchyard leaks just 0.5 t of SF₆, but $0.5 \times 23,500 = 11,750$ t CO₂-e — a tiny leak that out-weighs the entire dairy herd. Always write the formula, substitute, then give units (t CO₂-e).

THE DISTINCTION THAT SEPARATES A'S FROM C'S

07 Potency is not the same as contribution

Here is the resolution to the CO₂ paradox, and the single most-tested idea in this dot point. A gas's warming potential (GWP) is its potency per kilogram. Its actual contribution to warming depends on potency multiplied by how much is emitted.

SF₆ is fantastically potent but humans release only a trickle of it, so its total contribution is tiny. CO₂ is feeble per kilogram, but we pour out tens of billions of tonnes a year, so it dominates real-world warming. Toggle the two views below.

⚠ **WATCH OUT**

The classic wrong answer “SF₆ has the highest GWP, so it is the biggest cause of climate change.” False. High GWP ≠ high contribution. The biggest cause is CO₂, because of the sheer quantity emitted. State both halves — potency and amount — to earn full marks.

THREE ANALOGIES THAT STICK

08 Ways to keep it straight in the exam

◆ ANALOGY

◆ Currency exchange GWP is an exchange rate and CO₂-e is the common currency. Just as you convert yen, euros and dollars into one currency to add up a travel budget, you convert CH₄, N₂O and SF₆ into CO₂-e to add up an emissions budget. The "rate" for methane is 28-to-1.

◆ ANALOGY

◆ Chilli vs rice Potency vs quantity. One chilli (SF₆) is far hotter than one grain of rice (CO₂). But eat a whole kilogram of rice and a single chilli, and the rice delivers more total heat to your mouth. Per-unit strength and total amount are different questions.

◆ ANALOGY

◆ The blanket and its thickness Think of each gas as a blanket over the planet. Radiative efficiency is how thick the blanket is; lifetime is how long it stays on the bed. GWP measures the total warmth the blanket delivers across its time on the bed, compared with a standard CO₂ blanket.

DECODE THE COMMAND WORD

10 What the verb is really asking for

◆ COMMAND WORDS

Most lost marks come from answering the wrong question. Tap a command word to see exactly what an examiner expects when it appears on a warming-potential item.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

STRUCTURE A WRITTEN ANSWER

11 The P·E·L method



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

For "explain", "compare" and "justify" items, a tidy structure wins marks the content alone won't. P·E·L = Point (state the idea), Example/Evidence (a number, gas or Cape Wirreanda detail), Link (tie back to the question).

NAMED EXAM TRAPS

12 Six ways students throw away marks

WATCH OUT

Trap 1 — "GWP = how much is emitted" GWP is potency per kilogram, independent of how much is released. Quantity is a separate factor.

WATCH OUT

Trap 2 — forgetting the benchmark Always anchor to $\text{CO}_2 = 1$. "Methane's GWP is 28" only means anything relative to CO_2 .

WATCH OUT

Trap 3 — ignoring the time horizon A GWP number is meaningless without a horizon. Methane is ~80 over 20 years but ~28 over 100.

WATCH OUT

Trap 4 — only naming one factor "Explain GWP" wants both infrared absorption and atmospheric lifetime, not just one.

WATCH OUT

Trap 5 — confusing GWP with ozone depletion CFCs damage ozone and have high GWP — but these are different properties. Don't blur them.

WATCH OUT

Trap 6 — dropping the units in $\text{CO}_2\text{-e}$ A bare number isn't an answer. Mass $\times$ GWP gives tonnes $\text{CO}_2\text{-e}$ (or kg $\text{CO}_2\text{-e}$). State them.

PRACTICE EXAM QUESTIONS

13 Five graded questions · 24 marks

★ PRACTICE & SELF-MARK

Attempt each in your logbook first, then reveal the weak answer, the strong answer and the marking guide. Mark your own attempt out of the total — your score feeds the dock at the bottom of the page.

★ ONE-SCREEN RECAP

KK06 cheat sheet

- ► GWP = heat trapped by 1 kg of a gas over its lifetime, relative to 1 kg of CO₂, over a set time horizon. Unitless. CO₂ = 1.
- ► Two factors: (1) infrared absorption strength per molecule, (2) atmospheric lifetime. High in either → high GWP.
- ► Time horizon matters: short-lived methane ≈ 80 over 20 yr but ≈ 28 over 100 yr; long-lived gases barely change.
- ► Rough values (100 yr): CO₂ 1 · CH₄ 28 · N₂O 265 · HFCs ~1,300 · SF₆ ~23,500.
- ► CO₂-e = mass × GWP. The common currency for adding different gases together. Always give units (t CO₂-e).
- ► Potency ≠ contribution. CO₂ has GWP 1 but causes the most warming (huge quantity); SF₆ has huge GWP but tiny contribution (tiny quantity).

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.